



Education

Zhejiang University	Ph.D., Earth Climate and Environment	Electromagnetic scattering, remote sensing and data assimilation	2022.09-2027.06
Zhejiang University	B.Sc., Atmospheric Science	GPA 4.6/5.0 (Rank: 1/15)	2018.09-2022.06

Profile

Research in particle electromagnetic scattering and atmospheric remote sensing, combining AI with physics-based computation. Focus areas include physical modeling and optical calculations for nonspherical and inhomogeneous atmospheric particles, radiative transfer, and satellite data assimilation for numerical weather prediction. Cross-disciplinary experience spans scattering algorithm development and satellite applications, with strong project execution skills. A solid foundation in mathematics and physics supports AI algorithm development for scientific computing, remote-sensing retrievals, and data assimilation.

Projects and Research Experience

10 SCI-indexed papers: 5 first-author and 1 second-author (advisor first); 2 additional manuscripts under review.
1 granted invention patent.

1. All-sky hyperspectral infrared satellite assimilation for numerical weather prediction

- 2023-2025:** Research collaboration at the CMA Earth System Modeling and Prediction Centre on nonspherical cloud-particle scattering and all-sky infrared satellite data assimilation.
- All-sky hyperspectral infrared assimilation:** Developed algorithms to assimilate FY-4B/GIIRS all-sky radiances in the CMA-GFS 4D-Var system. (Q. J. R. Meteorol. Soc., 2026, 152(778), e70185, JCR Q2, **first author**.) The satellite instrument bias-correction method received an invention patent (second inventor; collaborating advisor first).
- Assimilation and forecast impact:** Used single-observation, typhoon-case and cycling experiments to examine cloud effects on observation sensitivity and water-vapor information extraction, and assess improvements in humidity analyses, typhoon tracks and medium-range forecasts. (Q. J. R. Meteorol. Soc., 2026, 152(778), e70184, JCR Q2, **first author**.)

Projects: National Key R&D Program (2022-2025); NSFC Joint Funds (2025-2028).

2. AI-based atmospheric profile retrieval and 3D cloud reconstruction

- Bayesian and machine-learning multivariate retrieval and assimilation:** Combined Bayesian methods with Transformer-ResNet networks to retrieve temperature, humidity and cloud profiles accurately; assimilation improved short-range severe-convection forecasts. (Adv. Atmos. Sci., 2026, 43(2), 373-389, JCR Q1, **first author**; J. Meteorol. Res., 2026, JCR Q2, fourth author and first student author.)
- 3D cloud reconstruction from satellite and model data:** Fused FY-4B AGRI observations and CMA-MESO backgrounds in a Bayesian framework to reconstruct 3D cloud liquid water, ice water and cloud fraction. Evaluated reconstruction and assimilation impacts on precipitation forecasts in a squall-line experiment. (Weather Forecast., 2026, JCR Q2, third author; collaborating advisor second.)

Projects: Zhejiang Institute of Meteorological Sciences collaboration (2023); CMA rapid cloud-scattering computation collaboration (2022-2025).

3. Light scattering by nonspherical atmospheric particles and lidar remote sensing

- Dust lidar ratio-depolarization ratio relationship:** Used superellipsoidal particles and the invariant imbedding T-matrix method to study effects of shape, size and refractive index; model results agreed well with high-spectral-resolution lidar observations. (J. Geophys. Res. Atmos., 2022, 127(4), e2021JD035629, **first author**.)
- Uncertainty in measured dust refractive indices and derived optical properties:** Developed theoretical analyses and numerical experiments reproducing laboratory retrievals to quantify effects of nonsphericity, internal inhomogeneity and particle-size measurement biases. (Atmos. Chem. Phys., 2024, 24, 6911-6935, JCR Q1, **first author**.)
- Comprehensive atmospheric single-scattering optical database:** Built a database for nonspherical dust, black carbon, sea salt and ice crystals, validated models against multiple observation sources, and provided neural networks for fast, accurate optical-property prediction. (Bull. Amer. Meteor. Soc., 2026, 107(7), E1563-E1578, JCR Q1, **second author; advisor first**.)

Projects: NSFC Excellent Young Scientists Fund (2021-2023); Chinese Academy of Sciences Major Science and Technology Infrastructure Open-Sharing Project (2023-2026).

Honors and Awards

First Prize, National College Mathematics Competition (Non-Mathematics Category) (2019)

Second Prize, Zhejiang Physics Innovation (Theory) Competition (2019)

Top 100, Research China; Team Third Prize, Policy China (Team Leader) (2019, 2020)

Outstanding Graduate, Zhejiang Province and Zhejiang University (2022)

National Scholarship (2021)

Selected for Zhejiang University Doctoral Qushi Rising Star Program (2024)

Zhejiang University First-Class Scholarship (2021, 2024, 2025)

Zhejiang University Outstanding Youth League Cadre / Member / Graduate Student (2020/21/23/24)